CLI and DOS Experience + Tasks

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1. Introduction

- Purpose of using Command Line Interface:

The Command Line interface (CLI) provides a faster way to relay instructions to the computer as opposed to clicking through a Graphical User Interface (GUI). This makes it much easier and less time intensive to perform repititive tasks and automate some other kinds of operations.

1.1 Commands shown in Class

```
# Win + R -> cmd
dir
#Listings special: cd . -> Current folder, cd . . -> Parent
 ↪  folder
#3rd fn of cd -> Shows name of current folder
mkdir foldername
#creates a new directory
rmdir foldername
#deletes a directory (has to be empty unless you force it)
move file_name1 destination
#moves a file to another location, same syntax logic as copy
ren old_name new_name
#renames a file
del file_name
#deletes a file
cd | clip
#piping, copying the folder name to clipboard
start .
#opens folder in which you are in
/?
#help
#no touch in Windows, directly echo smth to get the file out
 ↪  of nowhere
>
#overwrite
>>
#append
type file_name
#views file in cmd
copy file_name1 file_name2
```

```
#even when second file DNE
copy ..\a\*.txt ..\c
#everything in one line when you are in folder b
/p
#pauses output aka asks y/n every step e.g., deletion
/s
#forces, force del like when you have smth in the dir you want
 ↪  to remove
wd -d .
#go to the designated folder in GUI and type this on the
 ↪  window to directly access Powershell
cmd
#same way, cmd also words and cmd is ultimate destination even
 ↪  from Powershell
copy /filepath .
#copies to this folder without creation of new name
magick clipboard: cli.png
#first use imagemagick dwld and then use snipping tool
cls
#clears the clutter on screen
```

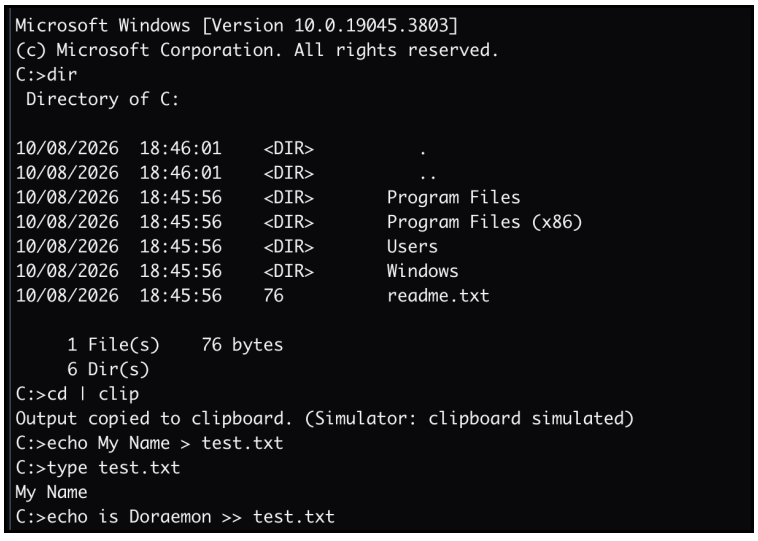


Figure 1: Example Output

Note: Since I lack a Windows device, a online simulator has been used.

1.2 Commands in Factsheet

```
del /p
#same /p as before but specifically for del -- confirms Y/N
 ↪  before deleting ech file, not force delete
exit
#closes the terminal session
dir /w
#lists contents in wide format, squeezes more names per line
 ↪  so less detail
dir /b
#bare format (names only), no dates/sizes/etc
echo %cd%
#prints current folder path as text
more filename
#views file page by page, pauses at each screen
python
#launches Python interpreter if installed
wgnuplot
#launches Gnuplot for plotting if installed
where python
#finds the filepath of the installed python.exe
```

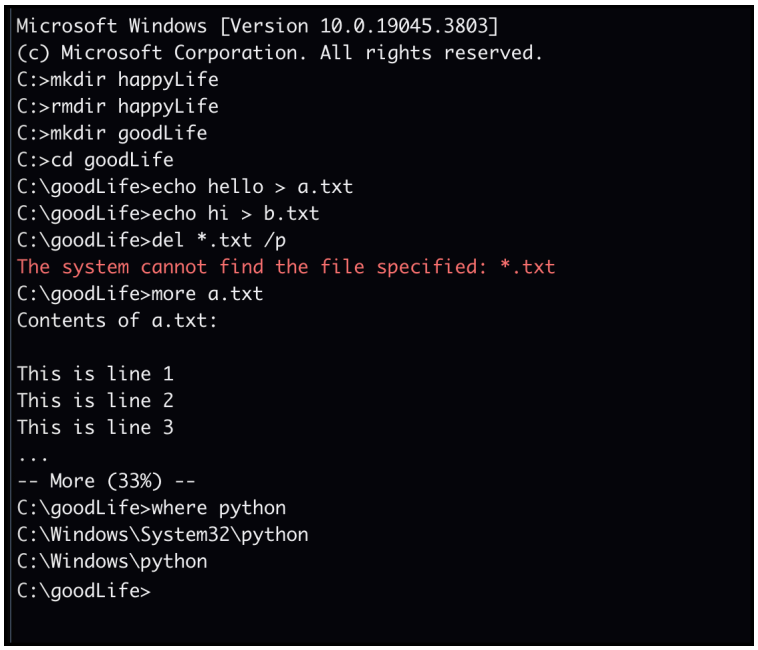


Figure 2: Sample Output on Simulator

Note: It is worth mentioning I couldn't run all the outputs on a simulator. However the tasks in section two were performed on a real Windows machine last week hence are realistic.

1.3 Introduction to Markdown

Markdown is the go-to method for digitally recording information for scientific purposes. The process of styling your documents is much simpler in `.md` files versus `.docx` files which is one of the reasons to prefer markdown. Multiple markdown editors exist in the market. To convert a markdown file to PDF one can use `pandoc` which can be installed using HomeBrew in MacOS.

```
**Apple**
#Bold
*Apple*
#Italics
#We are using marktext for md files
#In MarkText Ctrl+P to go to Editor View.

![ ](imgName.png ) #insertion of images
# ! means 'show me'

Hey this is an eqn $ inline eqn $

$$ in new para line to initiate a lone standing eqn then type
 ↪  your eqn:
```

To demondstrate the usage of these features, we were instructed to create markdown files in class (`apples.md`)

Notable equation example shown:

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r},t)\right]\Psi(\mathbf{r},t)$$

2. Experience (Workspace setup, MacOS)

Since I am on a machine running MacOS, the process of installation differs vastly from the one in Windows.

Firstly, the command-line package manager for MacOS isn't one distributed by Apple natively so you can use any of the free ones found online. I use `brew` and `brew install <package>` manages fetch/install/updates without manual intervention.

The packages notably mentioned in Lab to be installed included:

- pandoc
- Imagemagick
- GnuPlot
- Python

2.1 About Python

It is worth noting that instead of using the native `pip` globally, I prefer creating a virtual environment per project and locally install any module I need. This is a precaution against certain dependency conflicts (like different projects needing different versions) that I have faced in the past and a certain `error: externally-managed-environment` too.

2.2 About MarkText

This was the one suggested in the lab however the version for MacOS is depreciated and hence I switched to using `VSCode` with proper extensions. Also, worth mentioning, I used the `xelatex` renderer with pandoc since I had used that before and already had a compatible yaml template.



Figure 3: Depreciation Message

3. Commands and Outputs

3.1 Task 1: Directory creation and verification

```
cd PHY134SEM12026
mkdir CLI_DOS_July2026
cd CLI_DOS_July2026 dir /p
```

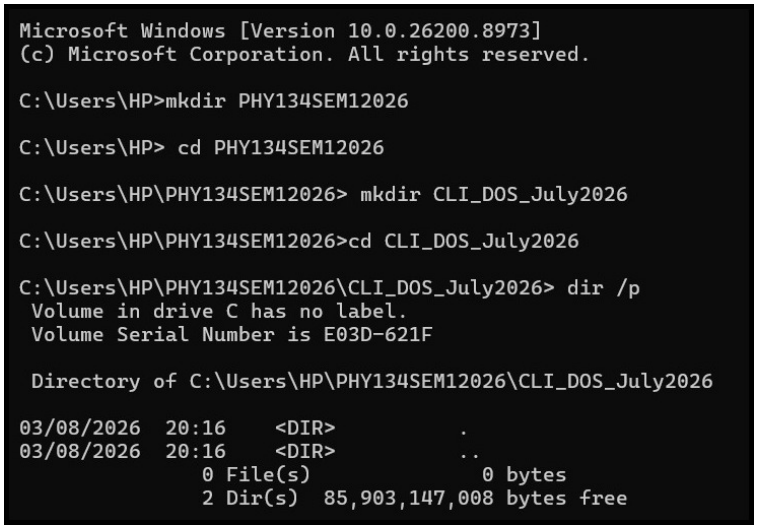


Figure 4: Task 1

3.2 Task 2: File creation

```
echo CLI DOS Experiment Notes > notes.txt
type notes.txt
more notes.txt
```

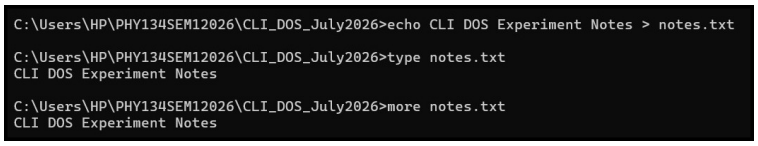


Figure 5: Task 2

3.3 Task 3: File Relocation

```
mkdir Data move notes.txt Data
cd Data dir
cd ..
```

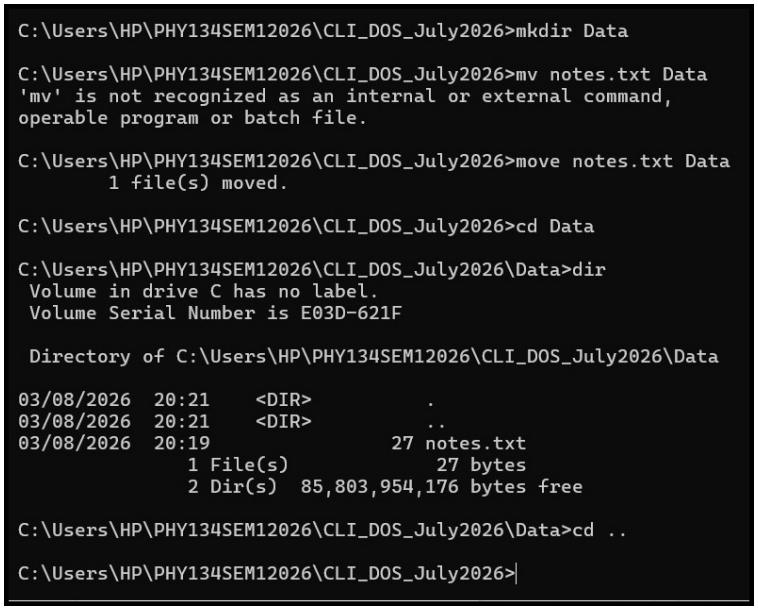


Figure 6: Task 3

3.4 Task 4: System Diagnostics Script

```
type system_report.txt
```



Figure 7: Task 4

3.5 Task 5: Txt File Creation and update

```
echo Test File > test.txt
echo Updated on %DATE% >> test.txt
cls
```

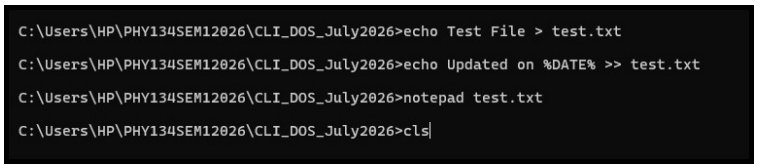


Figure 8: Task 5

3.6 Task 6: sin(x) Plotting

```
cd gnuplot
"C:\Program Files\gnuplot\bin\gnuplot.exe"
plot sin(x)
```

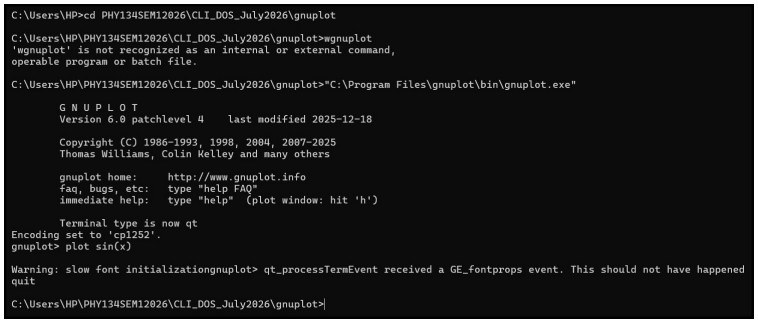


Figure 9: Task 6 - Script

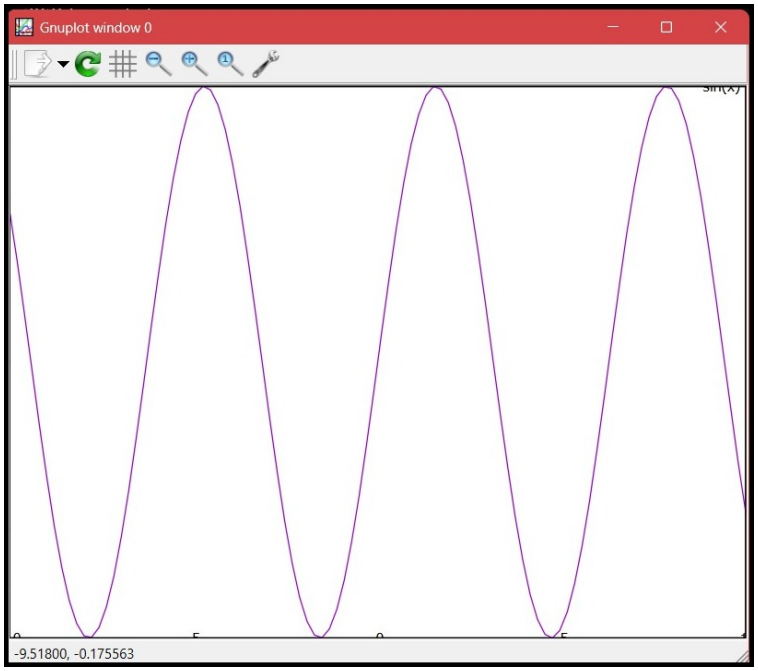


Figure 10: Task 6 - Plot

3.7 Task 7: Directory listing

```
dir /b > bare\_list.txt
more bare\_list.txt
```

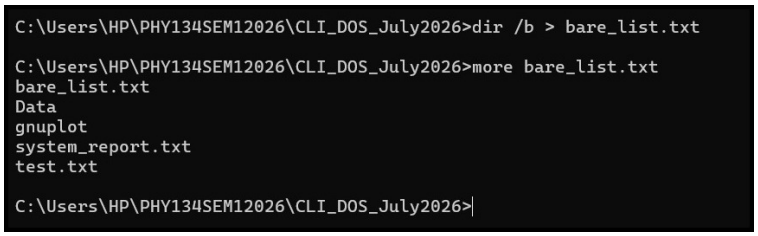


Figure 11: Task 7